SAAYA: AN IoT-ENABLED COOL SHELTER FOR OUTDOOR WORKERS

Project Report (concise version)

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Abstract. Outdoor workers in Vellore rest under bare metal sheets that become radiant heaters in the afternoon. Nobody measures whether these shades help, fans fail unnoticed and there is no reliable way to call for help. SAAYA is a demountable cool shelter for four to six workers: a 2.4 m × 2.0 m steel frame with a reflective upper roof skin, a ventilated 120 mm gap, a painted lining, a 200 mm overhang, a louvred screen and a bench. An ESP32-based node reads seat and outdoor air (two SHT30), roof lining temperature (DS18B20), fan speed (tachometer) and fan power (INA219); it runs a 12 V PWM fan only when moving air helps, stops automatic operation at 39 °C, detects fan faults and latches help requests. A screening heat model predicts that shade reduces mean radiant temperature at the seat by 15.9 °C and the cool roof by a further 2.6 °C, with the lining 26 °C cooler than a weathered sheet. The node costs ₹5,314 and draws 3.2 W. Firmware compiles and passes automated tests; fabrication and field testing are next.

1. Introduction

Vellore's mean daily maximum in May is about 39 °C, with a record of 45 °C [1]. The ILO projects that India will lose 5.8 % of working hours to heat stress by 2030, mostly in agriculture and construction [2]. Rest breaks are the main protection workers have, yet they are usually taken under galvanised or cement sheets whose undersides reach 60–80 °C at noon and radiate that heat onto people's heads. NDMA's draft guidance on cooling centres treats shaded rest areas as an essential first step [3].

Problem. Existing shades have three gaps: (1) the roof itself is hot; (2) there is no feedback on whether the shade or fan works, so a stalled fan looks the same as a running one; (3) a worker who feels unwell has no signal that cannot be missed.

Objectives. (a) Design a low-cost, demountable shelter that reduces radiant heat by passive means; (b) build an IoT node that controls a fan safely, detects faults and provides a latched help request; (c) make every cooling claim depend on paired, repeated measurement; (d) keep the electronics under ₹6,000 and below 10 W.

2. Shelter Design



Fig. 1: SAAYA cool shelter (CadQuery 3D model)

The shelter is designed to block sun first and then to keep the surface above people's heads cool. Six layers are used in order of cost: **shade** (the largest single effect), a **white upper skin** that reflects most sunlight, a **ventilated gap** of about 120 mm through which wind carries away the heat the top skin absorbs, a **painted aluminium lining** that stays close to air temperature even as the white coat soils, a **200 mm overhang and removable timber jaali louvres** that cut low-angle sun while keeping breeze, and finally a **controlled fan**, the only part that uses electricity.

Table 1: Main dimensions and members of the shelter (from the 3D model)

Item	Value	Reason
Post centres / roof	2400 × 2000 mm / 2800 × 2400 mm	Bench for 4–6 people; 200 mm overhang widens the shadow
Clear height	2310 mm	Standing headroom with margin
Roof fall	120 mm over 2000 mm (3.4°)	Rain run-off to the rear
Roof gap	≈ 120 mm (SHS 30 × 30 spacers)	Enough for wind-driven ventilation
Posts / beams / purlins	SHS 60×60×3 / RHS 80×60×3 / RHS 60×40×3	Standard sections, locally fabricated
Base	200 × 200 × 10 plates, 4 anchors each	Bolted, so the shelter can move with the site
Bench	1400 × 450 mm timber slats, seat 450 mm	Drains and stays cooler than steel
Mass / parts	≈ 250 kg, 44 distinct parts, 143 instances	Clash check: zero intersections

Table 2: Material choices and rejected alternatives

Choice	Why	Rejected alternative
Steel SHS/RHS, bolted	Predictable strength, demountable, repairable	Bamboo (variable, termites); timber (rot); aluminium (cost, welding)
White upper skin	Reflects sunlight; cheap; repaintable	Bare GI: modelled underside ≈ 80 °C; polycarbonate transmits heat
Ventilated gap	Lining stays cool as paint ages	Sealed gap (hotter); sandwich panel (cost, stores heat into evening)
Jaali louvres	Block low sun, keep airflow	Solid wall (blocks breeze); fabric (UV wear)
No AC / no air cooler	Open shelter cannot hold cooled air	AC 1.4–1.6 kW wasted; evaporative cooler weak in humid monsoon

3. Heat Model and Results

A screening model was written (api/thermal.py) to compare roof options before building anything. It balances solar gain, convection ($h = 2.8 + 3.0v$) [4], long-wave exchange with a clear sky [5] and view factors from the seated head (1.1 m) to roof, sky and ground, for Vellore at solar noon in mid-May (38 °C air, 45 % RH, 900 W/m², 1 m/s wind). Its output is the mean radiant temperature (MRT), the single number that captures how much heat surfaces radiate onto a person. The energy balance closes to zero and the model has 17 automated tests; it is a screening tool, not yet validated.

Table 3: Predicted MRT at the seated head and roof underside temperature

Design	MRT (°C)	Roof underside (°C)
Open sun, no shelter	73.6	—
Weathered GI / cement shed (typical today)	57.8	67.0
Single bare galvanised sheet	56.7	79.5
Single sheet, white both faces	55.4	48.9
SAAYA ventilated double roof	**55.2**	**41.0**

Findings. (1) Shade alone lowers MRT by 15.9 °C; this is the main benefit of any shelter. (2) The SAAYA roof adds a further 2.6 °C at the seat, because the roof fills only about 15 % of the seated person's view; but its underside is 26 °C cooler than a weathered sheet (41 vs 67 °C), which matters for standing workers and keeps working as the top coat soils. (3) Layout matters as much as the roof: +0.3 m of overhang gives −1.4 °C, irrigated ground cover around the shelter gives −3.9 °C, while light paving raises MRT by +4.8 °C by reflecting sun into the shelter. (4) The model makes falsifiable predictions: roof underside 40–45 °C and 150 mm globe 46–55 °C; it is rejected if measurements fall outside on two of three clear days.

Table 4: SAAYA compared with normal cooling options

Criterion	Tin shed	AC (1.5 t)	Air cooler	SAAYA
Power	None	1.4–1.6 kW	100–200 W + water	3.2 W typical
Works in an open shelter	Yes, hot roof	No	Partly	Yes
Humid monsoon	—	Works	Weak, adds humidity	Works
Knows if it is working	No	No	No	Yes (paired sensors)
Fan fault / help request	No	No	No	Yes / latched
Equipment cost	Low	≈ ₹36,990 + wiring	₹8–15k	₹5,314 node

4. IoT System: Architecture and Control

The SAAYA Node has three blocks. **Sense:** SHT30 at the seat (1.1 m, in a radiation shield), SHT30 outside as a reference (≥ 2 m away, same height), DS18B20 bonded to the roof lining, INA219 on the 12 V fan supply, the fan tachometer, a float switch in the drinking-water can, and MODE, HELP and ACK buttons. **Decide:** an ESP32 runs the control loop on one core with the rules kept in a single header that is unit-tested on a PC; a Wi-Fi uplink task runs on the other core; a 5 s watchdog restarts the board if anything hangs; the help latch is stored in flash. **Act:** 12 V PWM fan, OLED display, LED and buzzer; one JSON record per second goes over USB or Wi-Fi to a FastAPI/SQLite logger. Control never depends on the network.

Table 5: Control rules implemented in firmware (controller.h)

Rule	Behaviour	Reason
Auto fan	ON at 50 % when seat air ≥ 32 °C; OFF below 31 °C	Hysteresis prevents on/off flicker
Heat cut-off	Caution ≥ 37 °C; auto stop ≥ 39 °C; restart < 38.5 °C	Fans can add heat in very hot air [6, 7]
Manual	MODE gives 70 %; never blocked, warning shown	The worker decides
Fan fault	Duty > 0 but < 100 rpm after 3 s → FAULT	A stalled fan must not be reported as cooling
Help	HELP latches LED, buzzer, OLED; saved in flash; only ACK (side) clears	Cannot be lost by reboot or accidental press
Sensor loss	Reading shown as unavailable, never as 0	Missing data is not evidence
Fail-safe	If the ESP32 stops, PWM floats and the fan runs at 100 %	4-wire fan standard [8]

5. Component Selection and Justification

Table 6: Components, reasons and rejected alternatives (prices: Indian online retail, Oct 2026)

Component	Why chosen	Rejected, and why	₹
ESP32-DevKitC (WROOM-32)	Wi-Fi, 16 PWM channels, interrupt input for tach, flash, watchdog, instant boot, −40 to 85 °C [9]	Arduino Uno: no Wi-Fi, 2 kB RAM. Raspberry Pi: SD card corrupts on power cuts, ~10× power	499
SHT30 × 2	±0.2 °C, ±2 %RH typ.; addresses 0x44/0x45 share one bus [10]	DHT22: slow, RH drift. BME280: self-heating. SHT31: 2× price, same accuracy in 0–65 °C	358
DS18B20 probe	Waterproof, 1 m cable, one GPIO, ±0.5 °C [11]	IR sensor: wrong on low-emissivity metal. NTC: needs calibration	49
INA219	Measures real fan voltage and current on I²C [12]	ACS712: ≈ 60 mV signal at 0.33 A, noisy. Datasheet watts: not a measurement	89
Arctic P12 Pro 4-wire fan	25 kHz PWM, tach 2 pulses/rev, fails ON if PWM lost, 0.33 A	2-wire fan: no feedback. 230 V pedestal fan: shock risk, 55–125 W	910
SSD1306 OLED	Readable without a phone; shares I²C (0x3C)	16×2 LCD: 5 V, 6 pins. Phone app: workers may not carry phones	130

Component	Why chosen	Rejected, and why	₹
3 × 12 mm IP65 buttons	Rain-proof; HELP front, ACK on side for staff	Touch sensors: false triggers from sweat and rain	477
SFM-27 buzzer + LED	> 95 dB at < 12 mA	Passive piezo: quieter, needs tone generation	64
PP float switch	Dry contact; broken wire reads “low water” (fail-safe)	Ultrasonic: cost, condensation errors	175
12 V 2 A SMPS + MP1584 + 1 A fuse	12 V drives fan directly; efficient buck to 5 V	7805 linear: > 50 % lost as heat in a sealed box	257
ProtechT WPE16 IP65 ABS box	175 × 100 × 60 mm, rain-rated, purchasable	3D-printed PETG: softens at 60–70 °C, not rain-rated	949

Compatibility. All sensors and the OLED run at 3.3 V from the ESP32’s regulator, so no level shifting is needed; the fan’s PWM line is pulled down by a 2N3904 and its open-collector tachometer is pulled up to 3.3 V, so no GPIO ever sees 12 V. The four I²C devices have four different addresses (0x44, 0x45, 0x3C, 0x40) and the bus runs at 50 kHz for the longer reference cable. Input-only pins IO34/IO35 are used only as inputs with external 10 kΩ pull-ups; strapping pins (0, 2, 12, 15) and flash pins (6–11) are avoided. Peak current is 578 mA against the 2 A adapter, and the 3.3 V rail carries about 25 mA of its ≈ 500 mA budget. The 750 ms DS18B20 conversion runs asynchronously, so the loop is never blocked.

6. Interfacing Diagram

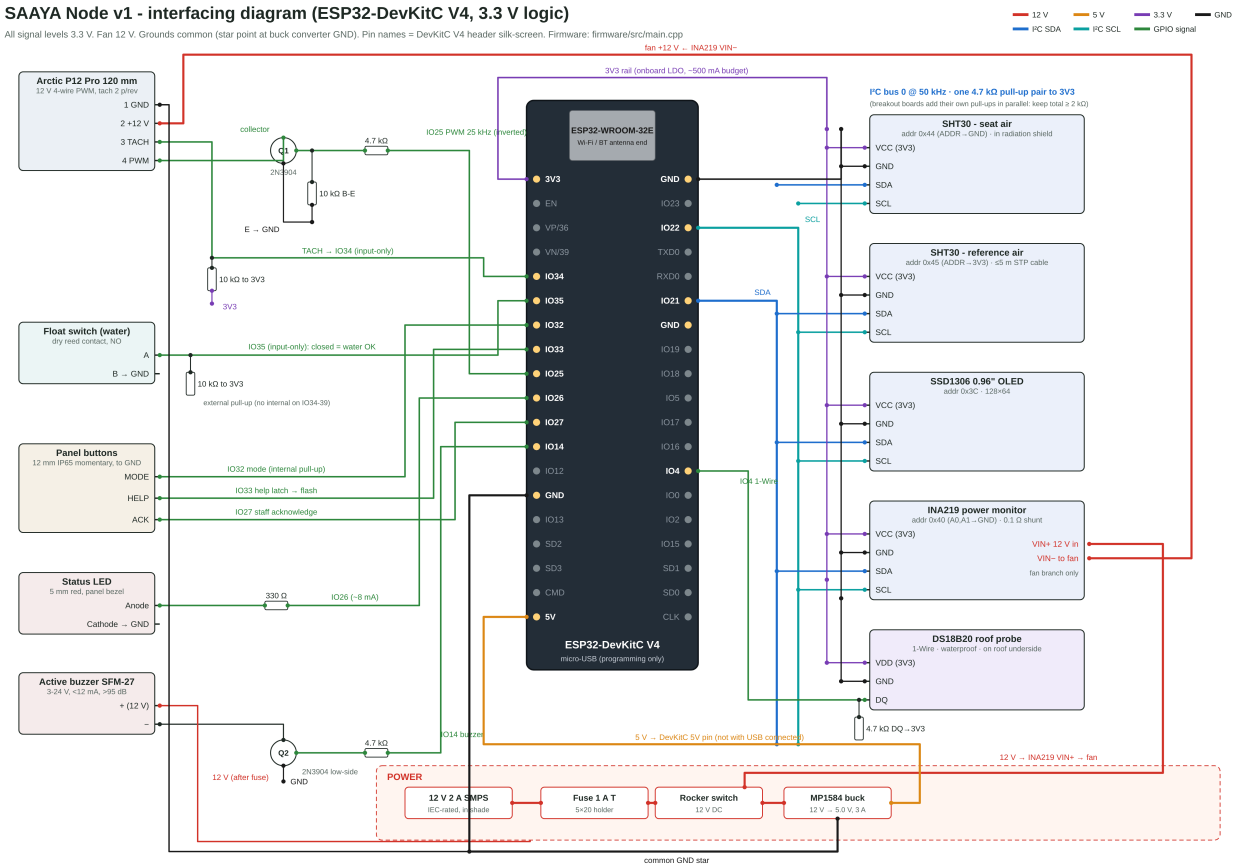


Fig. 2: SAAYA Node interfacing diagram (ESP32-DevKitC V4, 3.3 V logic)

Table 7: Pin assignment

Signal	ESP32 pin	Interface detail
I²C SDA / SCL	IO21 / IO22	SHT30 ×2, OLED, INA219; 4.7 kΩ pull-ups on modules; 50 kHz
Fan PWM	IO25	25 kHz via 2N3904 (inverted in firmware)

Signal	ESP32 pin	Interface detail
Fan tachometer	IO34 (input only)	10 kΩ to 3.3 V; interrupt, 2 pulses per revolution
Roof probe	IO4	1-Wire, 4.7 kΩ pull-up
Float switch	IO35 (input only)	10 kΩ pull-up; open = low water
MODE / HELP / ACK	IO32 / IO33 / IO27	To GND, internal pull-ups, debounced
LED / buzzer	IO26 / IO14	330 Ω (≈ 8 mA) / 2N3904 low-side switch on 12 V
Power	5V pin, GND	12 V → 1 A fuse → MP1584 (5 V) → on-board 3.3 V LDO; never USB and 5V pin together

7. 3D Model, Technical Specifications and Deployment

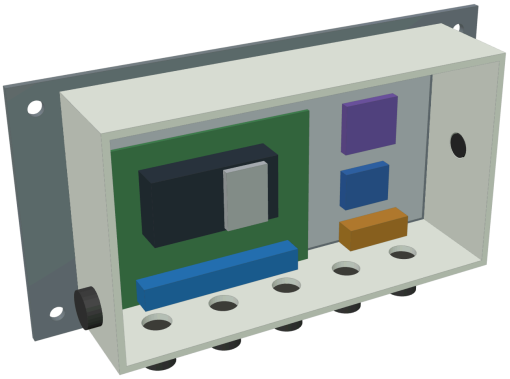


Fig. 3: Node enclosure, lid removed

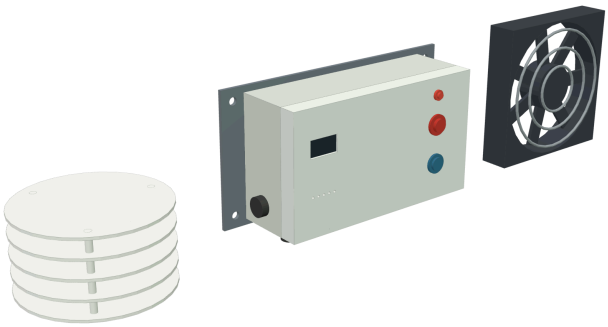


Fig. 4: Radiation shield, enclosure front and 120 mm fan

Table 8: Technical specifications of the SAAYA Node

Parameter	Value
Enclosure	175 × 100 × 60 mm IP65 ABS; 5 × PG7 glands on the bottom at 30 mm pitch; M12 breather vent
Front panel / side	OLED, HELP Ø12 mm, MODE Ø12 mm, LED Ø8 mm / ACK Ø12 mm on right wall
Mounting	3 mm aluminium bracket 215 × 110 mm on a 60 mm SHS post (hose clamps), box at 1.4 m
Supply / consumption	12 V DC 2 A; 3.2 W typical, 6.9 W peak, ≈ 34 Wh/day
Measurement	Air T/RH ±0.2 °C/±2 %RH (two points); roof ±0.5 °C; fan rpm, V, I
Sampling / output	1 Hz control; one JSON record per second over USB or Wi-Fi
Operating range	0–60 °C ambient inside the box; 5 s watchdog
Verification	Firmware compiles in CI (USB and Wi-Fi builds); native controller tests; 23-solid clash check

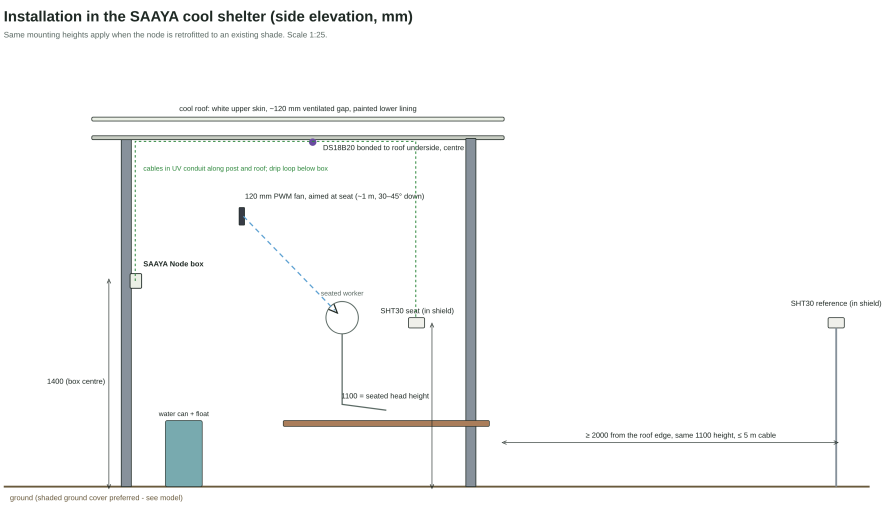


Fig. 5: Installation in the SAAYA cool shelter (side elevation, mm)

Deployment at a Vellore worksite. The open front faces away from the afternoon sun and the louvre screen is placed on the west side. The node is fixed to a rear post at 1.4 m with glands facing down and drip loops on every cable; the seat sensor hangs at 1.1 m in its shield; the reference sensor stands ≥ 2 m outside at the same height so that both see the same weather; the roof probe is bonded to the centre of the lining. Power comes from a 12 V mains adapter, or off-grid from a 20 W panel with a 12 V 7 Ah battery covering the ≈ 34 Wh daily need. On an existing shed the node can be retrofitted alone. In a typical day, the fan starts automatically around late morning when seat air crosses 32 °C, the OLED shows temperatures and fan state, the system stops automatic fan operation if seat air passes 39 °C, and a worker who feels unwell presses HELP, which keeps the light and buzzer on until a supervisor presses ACK.

8. Cost, Verification Plan and Conclusion

Cost. The node costs ₹5,314 including cables, shields and mounting (₹3,965 for listed parts). The first shelter is estimated at ₹94,134–₹170,771 including foundations, fabrication, transport and contingency; this is an estimate from the CAD quantities, not a quotation. Cost falls with repeated production of standard bolted sections, use of existing slabs, an optional upper skin where a clean white roof is maintained, ground cover before a larger roof, and retrofitting the node onto existing sheds.

Verification plan. No cooling claim is made from the design alone. The margin is declared before data is collected (1.0 °C for roof temperature, because two ±0.5 °C sensors may disagree by up to 1 °C; 0.2 m/s for air speed). Seat readings are always paired with the outdoor reference taken at the same moment. A single run is reported only as a point estimate; a result requires at least three valid runs on three different days, and a direction is claimed only if the whole 95 % confidence interval clears the margin. Runs with a fan or sensor fault are withheld automatically. The website's evidence lab demonstrates this with replayed data.

Limitations. The shelter has not yet been built or measured; the heat model is unvalidated; steel sections, foundations and wind loads require a structural engineer's approval; costs are estimates.

Conclusion. SAAYA combines a passive cool shelter, chosen with the help of a heat model, with a ₹5,314 IoT node that runs a fan only when it helps, stops at 39 °C, detects faults, latches help requests and refuses to claim cooling without paired, repeated measurement. The next steps are structural approval, building one shelter with two nodes, co-locating sensors to correct offsets, and a three-day paired test comparing single and double roofs.

Project files: github.com/alphasafal/saaya-verify · Live site and 3D models: saaya-verify.vercel.app

References

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